

searchnet: Network-Embedded Strategic Search Simulation in R

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Abstract

The NK fitness landscape model is among the most widely used computational frameworks in strategy and organization theory, yet it treats the complexity parameter K as exogenous, the landscape as fixed, and agents as isolated searchers. The R package **searchnet** addresses all three limitations by reformulating the NK model as a bipartite network of M actors affiliating with N components and embedding it within a Stochastic Actor-Oriented Model (SAOM). This reformulation reveals that the scalar K is a special case of a four-dimensional coupled degree system—actor scope (K_{AC}), component popularity (K_{CA}), actor sociality (K_{AA}), and component epistasis (K_{CC})—whose dynamics are governed by actors’ forward-looking affiliation decisions rather than imposed by the modeler. The package provides an R6-based simulation engine built on **RSiena**, a clean functional API (`saomnk_env`, `saomnk_model`, `saomnk_run`), coupled degree visualization through the $\{K\}$ -4 panel, utility decomposition diagnostics, exogenous shock infrastructure, and specification robustness tools (SAI, CFC, DGF). Three worked illustrations demonstrate reproduction of classical NK search dynamics, endogenous landscape formation with heterogeneous strategies, and quasi-experimental shock analysis. Benchmarks against **nkmodel** and raw **RSiena** quantify computational overhead and scalability. **searchnet** targets researchers in strategic management, organizational theory, innovation economics, and computational social science who study search, adaptation, and competitive dynamics on complex landscapes.

1 Introduction

The NK fitness landscape model [Kauffman and Levin, 1987, Kauffman and Weinberger, 1989] has become one of the most productive computational frameworks in strategic management and organization theory. Since Levinthal [1997] demonstrated that the NK model could illuminate the tension between exploration and exploitation on rugged performance landscapes, hundreds of studies have deployed it to investigate organizational adaptation, modular design, imitation, and competitive search [Baumann et al., 2019, Rivkin, 2000, Siggelkow and Rivkin, 2005].

The model’s appeal lies in its parsimony. Two parameters— N , the number of design choices or components, and K , the degree of epistatic interdependence among them—generate landscapes ranging from smooth single-peaked surfaces ($K = 0$) to highly rugged terrains ($K = N - 1$) populated by many local optima. This parsimony has also become the model’s principal liability. Four limitations recur across the literature.

First, the fitness landscape is **exogenous**: it is drawn once from a random distribution and does not respond to the positions that agents occupy. Firms competing on the same landscape experience identical payoff surfaces regardless of how many competitors share each component or how concentrated competitive attention has become. Second, K is a **scalar** set by the modeler. In empirical organizational settings, interdependence is neither uniform nor fixed; some components are tightly coupled while others operate nearly independently, and coupling structures change as actors reconfigure their systems. Third, **multi-agent extensions** typically treat agents as independent searchers navigating the same landscape, ignoring the social and competitive structure through which firms observe, imitate, and crowd one another. Fourth, the NK model provides **no statistical framework** for parameter estimation from empirical data, making it difficult to discipline simulations with observed organizational behavior.

This paper introduces **searchnet**, an R package that addresses all four limitations simultaneously. The package reformulates the NK model as a bipartite network of M actors (firms, agents, decision-makers) affiliating with N components (technologies, resources, markets, design choices). By embedding this bipartite

structure within a Stochastic Actor-Oriented Model [SAOM; Snijders, 1996, Snijders et al., 2010], the search environment becomes a network data generation process whose topology—and therefore whose fitness landscape—co-evolves with actors’ affiliation decisions.

This reformulation yields three contributions.

Endogenization. The fitness landscape is no longer exogenous. Landscape ruggedness emerges from the structure of the bipartite network, which is itself shaped by actors’ strategic choices. When multiple actors affiliate with the same components, they create epistatic coupling through overlapping activity systems; when they withdraw, coupling dissolves.

Decomposition. The scalar K is revealed as a special case of a four-dimensional coupled degree system, denoted $\{K\}$. The four dimensions—actor scope (K_{AC}), component popularity (K_{CA}), actor sociality (K_{AA}), and component epistasis (K_{CC})—are individually standard network statistics [Latapy et al., 2008, Borgatti and Everett, 1997], but the analytical leverage of the $\{K\}$ framework comes from treating them as a coupled system. Because all four derive from the same bipartite incidence matrix, a change in any one dimension mechanically propagates to the others, exposing the transmission mechanism through which micro-level affiliation decisions produce macro-level landscape topology.

Statistical infrastructure. Because the model is formalized as a SAOM, it inherits **RSiena**’s [Ripley et al., 2022] well-developed statistical machinery, including maximum likelihood estimation, goodness-of-fit testing, and the capacity to be estimated on empirical network panel data.

A full-scale empirical application using eight data layers from the U.S. airline industry is reported in Downing [2026].

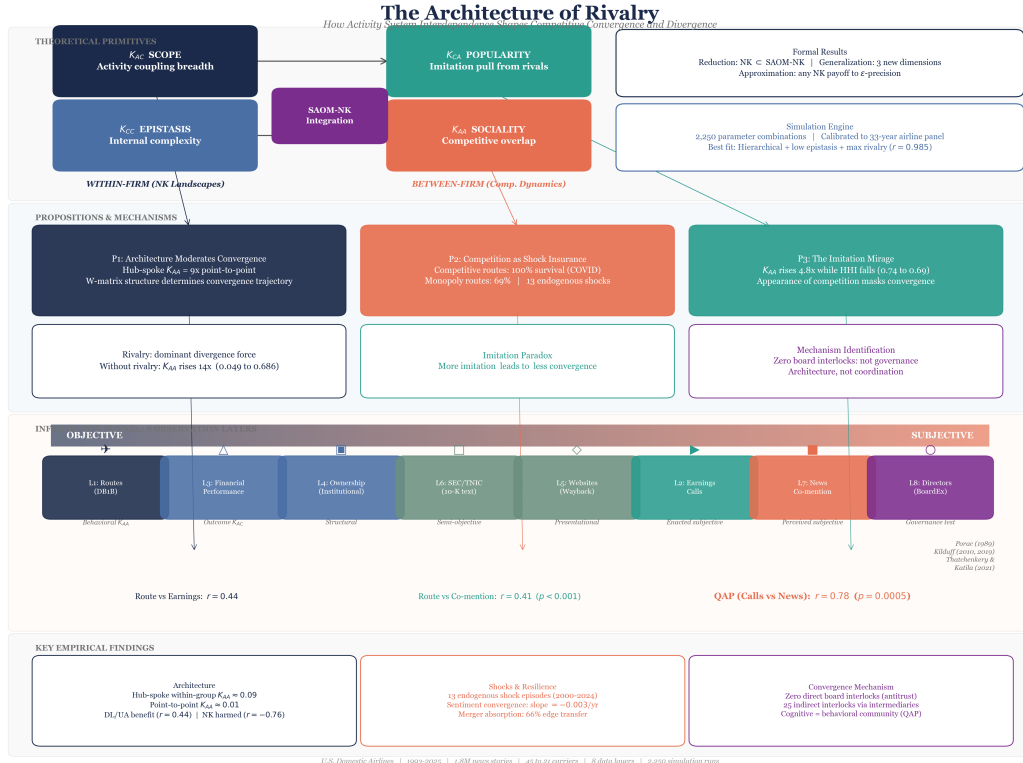


Figure 1: The $\{K\}$ Interdependence System: four coupled degree dimensions emerging from the bipartite actor-component network. Adapted from Downing (2026a).

The package provides an R6-based simulation engine, a functional user API that abstracts away **RSiena**’s internal naming conventions, comprehensive visualization through the $\{K\}$ -4 panel and 3D phase space,

utility decomposition diagnostics, exogenous shock infrastructure for quasi-experimental designs, multi-wave Monte Carlo facilities, specification robustness tools, an interactive strategy game mode, and a classroom teaching module. Eight vignettes accompany the package: a quick-start introduction, theoretical foundations, a full simulation walkthrough with strategies and shocks, a computational experiment design guide, a causal inference tutorial, an NK validation guide, a Blume–Gibbs convergence tutorial, and a Professional Development Workshop curriculum.

The remainder of this paper is organized as follows. Section~2 reviews the NK and SAOM literatures and positions the contribution. Section~3 presents the SaoMNK framework: the bipartite formulation, the $\{K\}$ interpretive framework, the SAOM objective function, and the rationale for choosing SAOMs over ERGMs. Section~4 describes the **searchnet** package architecture, user API, visualization tools, diagnostics, animation pipeline, interactive features, and proof registry. Section~5 provides three worked illustrations. Section~6 presents computational benchmarks and scalability analysis. Section~7 concludes.

2 Background: NK models and SAOMs

2.1 The NK fitness landscape

The NK model [Kauffman and Levin, 1987, Kauffman and Weinberger, 1989] characterizes performance landscapes with two parameters. A system of N binary components (loci, design choices, technologies) generates 2^N possible configurations. Each component d makes a fitness contribution f_d that depends not only on its own state but also on the states of K other components with which it is epistatically linked. Total fitness is the average contribution across all N components. When $K = 0$, each component contributes independently and the landscape has a single global peak. As K increases toward $N - 1$, the landscape becomes increasingly rugged, populated by exponentially many local optima that trap myopic searchers.

Levinthal [1997] introduced the NK model to management as a formalization of the exploration–exploitation trade-off: agents that search locally (single-bit mutations) on rugged landscapes quickly find nearby peaks but rarely discover the global optimum, while agents that search more broadly (multi-bit jumps) sacrifice short-run performance for a chance at higher peaks. This insight has been extended to organizational design [Siggelkow and Rivkin, 2005], imitation [Rivkin, 2000], modular architectures [Ethiraj and Levinthal, 2004], and competitive interaction [Lenox et al., 2006].

Despite its generativity, the classical NK model imposes structural constraints that limit its applicability to settings where search environments are dynamic, socially embedded, or empirically observable. The landscape is drawn once and held fixed; K is treated as a uniform scalar; agents search independently; and there is no estimation framework linking the model to data. Several extensions have relaxed individual constraints—Gavetti and Levinthal [2000] introduced cognitive search, Siggelkow and Rivkin [2005] embedded search in organizational hierarchies, and Baumann et al. [2019] provided a comprehensive review—but no existing framework addresses all four limitations simultaneously within a unified statistical model.

2.2 Stochastic Actor-Oriented Models

Stochastic Actor-Oriented Models [Snijders, 1996, Snijders et al., 2010] provide a continuous-time Markov model for the co-evolution of social networks and behavior. At each micro-step, a single actor is selected according to a rate function, evaluates the marginal utility of each possible tie change according to an objective function, and chooses probabilistically via a multinomial logit rule. The objective function takes the general form

$$f_i(\mathbf{B}) = \sum_k \theta_k s_{ik}(\mathbf{B}), \quad (1)$$

where $s_{ik}(\mathbf{B})$ are network statistics computed from the current network $\mathbf{B}$ and θ_k are parameters governing each actor’s preferences. The parameters θ_k can be estimated from observed panel data using the method of moments or maximum likelihood, implemented in the **RSiena** package [Ripley et al., 2022].

Table 1: The $\{K\}$ coupled degree framework. Each dimension is a standard network statistic; the contribution is their joint treatment as a coupled system.

Dimension	Symbol	Interpretation	Network measure
Actor scope	K_{AC}	Components per actor	Bipartite row degree
Component popularity	K_{CA}	Actors per component	Bipartite column degree
Actor sociality	K_{AA}	Co-affiliating actors per actor	Actor projection degree
Component epistasis	K_{CC}	Co-occurring components per component	Component projection degree

SAOMs are well established in sociology and political science for modeling friendship networks, advice networks, and inter-organizational partnerships. They have been extended to bipartite (two-mode) networks [Koskinen and Edling, 2012], where the model governs actors’ affiliation decisions with respect to events, organizations, or other categorical entities. **searchnet** exploits this bipartite extension to model actors affiliating with components in a search environment.

2.3 Positioning searchnet

No existing R package provides the capability that **searchnet** offers. The **RSiena** package [Ripley et al., 2022] implements SAOMs but does not include bipartite search environment simulation with NK-style fitness landscape analysis. The **nkmodel** package [Baumann, 2019] implements classical NK models but not the network-embedded generalization. **searchnet** bridges these two traditions, providing a simulation engine in which the fitness landscape is an emergent property of actors’ network positions rather than an exogenously imposed parameter.

3 The SaoMNK framework

Notation. Throughout this paper we adopt the notation of Downing [2026], where the formal SAOM-NK proofs are developed in full. The bipartite incidence matrix is $\mathbf{B} \in \{0, 1\}^{M \times N}$ with entries b_{ij} ; the NK configuration vector for a single agent is $\mathbf{x} \in \{0, 1\}^N$; the epistasis matrix is $\mathbf{E}$ (implemented in **RSiena** as the weight matrix $\mathbf{W}$ of the $\mathbf{XW}\mathbf{X}$ effect); component payoff functions are f_a ; and the logit precision parameter is $\beta > 0$.

3.1 Bipartite network formulation

The classical NK model represents a system configuration as an N -dimensional binary vector $\mathbf{x} = (x_1, x_2, \dots, x_N)$ where $x_j \in \{0, 1\}$. **searchnet** generalizes this by introducing M actors, each of whom maintains an affiliation vector. The resulting system state is an $M \times N$ binary matrix $\mathbf{B}$, where $b_{ij} = 1$ indicates that actor i affiliates with component j . This matrix is the incidence matrix of a bipartite network with actor nodes on one side and component nodes on the other.

The bipartite representation has three immediate consequences. First, it naturally accommodates **multiple agents** searching simultaneously: each row of $\mathbf{B}$ is an actor’s configuration, and actors’ configurations interact through shared columns. Second, it makes the **search environment endogenous**: the pattern of affiliations determines which components are coupled (via shared actors) and which actors are competitors (via shared components). Third, it provides the structural substrate for the SAOM engine: $\mathbf{B}$ becomes the dependent variable in an evolving bipartite network model.

3.2 The $\{K\}$ interpretive framework

The bipartite incidence matrix $\mathbf{B}$ generates four coupled degree statistics, each of which is individually a standard network measure but whose joint dynamics expose the mechanism through which micro-level search decisions produce macro-level landscape structure. Table~1 summarizes the four dimensions.

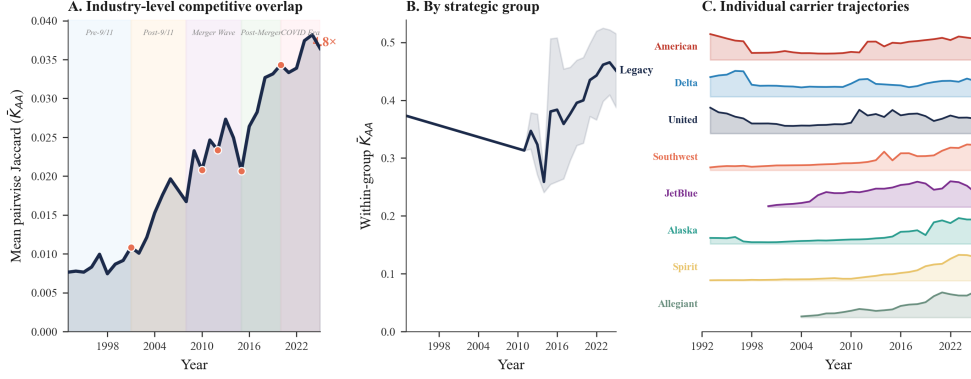


Figure 2: Empirical validation: K_{AA} (competitive overlap) among U.S. domestic airlines, 1993–2025 (Downing 2026a). The 4.8-fold increase in pairwise overlap demonstrates the $\{K\}$ system’s capacity to track competitive convergence dynamics using real network panel data.

The classical NK parameter K maps directly onto K_{CC} —the degree of a node in the component–component projection network. Two components share a projection edge whenever at least one actor affiliates with both, and the number of such edges incident on a focal component determines how many other components epistatically affect its fitness contribution. In the classical model, this coupling structure is set exogenously by the modeler; in **searchnet**, it emerges endogenously from the bipartite network and co-evolves with the other three degree dimensions.

The analytical leverage of the $\{K\}$ framework comes from treating these four measures as a coupled system rather than examining any one in isolation. Because all four derive from the same incidence matrix $\mathbf{B}$, they are structurally linked. A change in actor scope (K_{AC}) mechanically alters component popularity (K_{CA}), which in turn reshapes both the actor co-membership network (K_{AA}) and the component co-occurrence network (K_{CC}). The projections are computed as:

$$\mathbf{P}_{AA} = \mathbf{B}\mathbf{B}^\top - \text{diag}(\mathbf{B}\mathbf{B}^\top), \quad (2)$$

$$\mathbf{P}_{CC} = \mathbf{B}^\top\mathbf{B} - \text{diag}(\mathbf{B}^\top\mathbf{B}). \quad (3)$$

The degree of actor i in the actor projection $\mathbf{P}_{AA}$ is $K_{AA,i} = \sum_{h \neq i} \mathbf{1}[(\mathbf{B}\mathbf{B}^\top)_{ih} > 0]$, and symmetrically for $K_{CC,j}$ in the component projection. Analyzing any single measure in isolation obscures the cross-domain dependencies that link actor behavior to landscape topology.

It is important to be precise about what the $\{K\}$ framework contributes. The four statistics are individually well known in the bipartite network literature [Latapy et al., 2008, Borgatti and Everett, 1997]. The contribution is not statistical but **interpretive**: $\{K\}$ provides a theoretical bridge between the NK landscape tradition in strategy—where K is a scalar complexity parameter—and the bipartite network tradition in sociology—where degree distributions are descriptive summaries of affiliation structure. By grouping these four measures and tracking their joint evolution within a SAOM, the framework makes visible the transmission mechanism through which actors’ strategic choices propagate into landscape ruggedness.

3.3 SAOM objective function specification

The SAOM objective function in **searchnet** governs actors’ affiliation decisions. At each micro-step, the selected actor i evaluates the change in objective function value for each possible tie change (creating or dissolving an affiliation with one component) and selects the change that maximizes expected utility, subject to a stochastic error term drawn from the type I extreme value distribution.

The objective function takes the form of Equation~1, with the following key statistics:

Table 2: SAOM objective function statistics in **searchnet**.

Statistic	RSiena name	Formula	Interpretation
Density	density	$\sum_j b_{ij}$	Baseline affiliation rate
Popularity	inPop	$\sum_j b_{ij} \sqrt{\sum_{h \neq i} b_{hj}}$	Preference for popular components
Scope	outAct	$\sum_j b_{ij} \sqrt{\sum_{k \neq j} b_{ik}}$	Scope expansion tendency
Epistasis	XWX	$\sum_j b_{ij} \sum_k w_{jk} b_{ik}$	Exogenous component coupling

The density effect serves as an intercept: a negative density parameter creates baseline sparsity pressure, preventing actors from affiliating with all components. The popularity effect (**inPop**) creates preferential attachment to components that many other actors already use. The scope effect (**outAct**) governs actors’ tendency to expand or contract the breadth of their affiliations. The epistasis effect (**XWX**) introduces exogenous component–component coupling via a modeler-supplied weight matrix **W** (denoted **E** in the formal proofs; see Section~3.4.1); actors derive utility from affiliating with components that are coupled to their existing affiliations.

Actor-level heterogeneity enters through constant covariates attached to the objective function. A covariate vector **v** (e.g., encoding strategic orientation) can modify the weight an actor places on popularity seeking (**inPopX**), scope expansion (**egoX**), or component characteristics (**altX**). This mechanism allows the modeler to assign distinct strategic profiles to different actors—for example, “exploiters” who preferentially attach to popular components and “explorers” who expand into uncrowded niches.

The simulation proceeds via **RSiena**’s continuous-time Markov chain algorithm. Actors are selected according to a rate function (by default, equal rates for all actors), and the selected actor evaluates all possible single-tie changes, choosing according to a multinomial logit rule:

$$P(b_{ij} \rightarrow 1 - b_{ij}) = \frac{\exp(f_i(\mathbf{B}^{ij}))}{\sum_{j'} \exp(f_i(\mathbf{B}^{ij'}))}, \quad (4)$$

where $\mathbf{B}^{ij}$ denotes the network configuration obtained by toggling actor i ’s tie to component j . The **search_rsiena** method passes the full structure model to **RSiena**’s **siena07** function, which executes the continuous-time simulation internally. The **iterations_per_actor** parameter controls the total number of micro-steps: $M \times \text{iterations_per_actor}$.

For exogenous shocks, **searchnet** partitions the simulation timeline into proportional segments. At each segment boundary, the θ parameters are updated to new values specified by the shock schedule. The simulation then continues from the current network state with the modified objective function, preserving path dependence while allowing the researcher to model environmental discontinuities. This is equivalent to running sequential **RSiena** simulations where each wave’s initial state is the terminal state of the previous wave.

3.4 Mapping RSiena effects to strategic search concepts

The preceding objective function uses **RSiena** effect names that are standard in social network analysis but may be unfamiliar to strategy researchers. Table~3 provides a systematic mapping between **RSiena** effects available in **searchnet** and their strategic search interpretations. Each effect’s parameter symbol links it to the corresponding $\{K\}$ dimension introduced in Section~3.2, enabling researchers to translate between the network estimation vocabulary and the strategy-theoretic language of scope, complementarity, and imitation.

The table makes explicit the division of labor among effects. The first four rows—**density**, **outAct**, **inPop**, and **XWX**—are the workhorses of most **searchnet** specifications. Together they endogenize the four $\{K\}$ dimensions: scope cost sets the baseline, **outAct** introduces diminishing returns to breadth, **inPop** creates demand-side herding toward popular components, and **XWX** rewards configurations that exploit exogenous

Table 3: Mapping RSiena effects to strategic search concepts. Parameters correspond to the SAOM objective function (Equation 1). The $\{K\}$ column indicates which coupled degree dimension each effect primarily governs.

RSiena Effect	Parameter	$\{K\}$ Dim.	Strategy Concept	Interpretation
density	θ_{dens}	—	Scope cost	Negative = costly to hold activities; controls po
outAct	θ_{scope}	K_{AC}	Diminishing returns	Quadratic penalty on actor degree (scope)
inPop	θ_{pop}	K_{CA}	Bandwagon / legitimacy	Positive = prefer popular components; preferen
XWX	θ_{epist}	K_{CC}	Complementarity / synergy	Reward for holding epistatically linked compone
simEgoInDist2	θ_{imit}	K_{AA}	Imitation / herding	Positive = actors converge to similar portfolios
egoX	θ_{strat}	—	Strategy heterogeneity	Actor-level covariate modifying search behavior
altX	θ_{value}	—	Component value	Exogenous activity-level payoff differences
cycle4	θ_{close}	K_{AA}	Structural closure	Triadic closure in the projected social network

complementarities. The remaining effects allow the researcher to layer on additional mechanisms: imitation (**simEgoInDist2**) induces portfolio convergence among structurally similar actors, actor and component covariates (**egoX**, **altX**) introduce heterogeneity, and **cycle4** captures closure in the actor–actor projection of the bipartite network. By selecting which effects to include and setting their parameter values, the researcher controls which strategic mechanisms are active in a given simulation—a design choice whose consequences can be audited with the diagnostic tools described in Section~4.4.

3.4.1 Formal properties

The SaoMNK framework rests on formal results that establish its relationship to the classical NK model and to quantal response equilibrium. We state the main results here; complete proofs with all lemmas are provided in the Online Appendix [see also Downing, 2026] (available as `inst/proofs/` in the package; access via `searchnet_proof()`).

Theorem 1 (Reduction). The standard NK model is a degenerate special case of the SAOM-NK model. Specifically, the NK landscape and its adaptive walk are recovered exactly under five restrictions: (R1) $M = 1$ (single actor); (R2) $\mathbf{E} = \mathbf{A}$ (SAOM-NK epistasis matrix equals the NK interaction matrix); (R3) $\boldsymbol{\theta} = \mathbf{0}$ (all SAOM parameters zero); (R4) $\beta \rightarrow \infty$ (deterministic greedy choice); (R5) uniform component selection probability.

Theorem 2 (Generalization). SAOM-NK strictly generalizes the NK model along three independent dimensions: (a) multi-actor endogenous interaction ($M > 1$), (b) endogenous landscape co-evolution through SAOM network statistics, and (c) bounded rationality with tunable search precision ($\beta \in (0, \infty)$). The coupled degree vector $\{K\} = (K_{AC}, K_{CA}, K_{AA}, K_{CC})$ spans a strictly larger class of fitness landscapes than the scalar K parameterization. The classical NK model is recovered as a proper special case: $\text{NK} \subsetneq \text{SAOM-NK}$.

Theorem 3 (Approximation). For any NK landscape $(N, K, \mathbf{A}, \{f_d\}_{d=1}^N)$, there exists a parameterization of the SAOM-NK model that reproduces the NK payoff structure to arbitrary precision. Three independent proof strategies establish this: (a) constructive existence via dummy covariates, (b) actor heterogeneity substitution, and (c) universal approximation via mixed logit.

Theorem 4 (SAOM–QRE Equivalence). The SAOM multinomial logit choice rule (Equation 4) is isomorphic to a quantal response equilibrium (QRE) in the bipartite affiliation game, establishing a game-theoretic foundation for the statistical model.

Theorems~1–3 correspond to Theorems~1–3 in Downing [2026]; Theorem~4 is specific to this paper. These results are applied empirically to U.S. airline competitive dynamics in Downing [2026].

3.5 Why SAOM over ERGM?

A natural question is why **searchnet** adopts the SAOM rather than the exponential random graph model [ERGM; Lusher et al., 2013] or its temporal extension [TERGM; Hanneke et al., 2010] as its network engine. The choice follows directly from the theoretical premise of the NK tradition.

The NK model is fundamentally about **agents making strategic decisions**: an actor evaluates alternative configurations and selects the one that maximizes a fitness criterion. The SAOM’s rate-function architecture is the natural formalization of this process. At each micro-step, a single actor is selected, evaluates the marginal utility of each possible tie change, and chooses probabilistically according to a multinomial logit rule. This actor-centric, sequential-decision structure maps directly onto the local search process in NK models, where an agent evaluates neighboring configurations and moves to the best alternative.

ERGMs, by contrast, characterize the equilibrium distribution of network configurations without specifying a generative decision process. An ERGM answers the question “what network structures are over-represented relative to chance?” but does not model *who* makes *which* decision *when*. For a framework whose purpose is to endogenize landscape ruggedness through actors’ strategic choices, this equilibrium focus omits precisely the mechanism of interest.

Beyond theoretical alignment, SAOMs offer practical advantages for the simulation-based workflow that **searchnet** supports. The SAOM’s micro-step chain produces a complete trajectory of network states, enabling the researcher to track how $\{K\}$ dimensions evolve over time, to identify critical transitions, and to introduce shocks at precise points in the simulation. ERGM simulation, by contrast, generates draws from a stationary distribution without temporal ordering, making path-dependent analysis infeasible.

Finally, the SAOM’s continuous-time formulation handles bipartite networks with actor-level covariates naturally through **RSiena**’s existing infrastructure. While bipartite ERGMs exist [Wang and Pattison, 2009], the SAOM’s treatment of bipartite networks as actor-driven affiliation processes is better suited to the search-theoretic interpretation that motivates **searchnet**.

4 The searchnet package

4.1 Architecture

searchnet is built on R6 classes organized in a two-level inheritance hierarchy:

SaomNkRSienaBiEnv_base Core environment initialization, bipartite matrix management, graph construction, projection computation, and utility helpers (toggle, Jaccard, existence checks). This base class manages the $M \times N$ incidence matrix, constructs the **RSiena** data objects, and provides low-level network operations.

SaomNkRSienaBiEnv Full simulation engine inheriting from the base class, adding the `search_rsiena()` simulation method, chain statistics processing, $\{K\}$ -4 degree computation, fitness landscape evaluation, shock handling, multi-wave experiment infrastructure, and the visualization interface.

Supporting modules provide separation of concerns:

- **saomnk-api.R**: User-facing functional API (`saomnk_env`, `saomnk_model`, `saomnk_run`) that wraps the R6 internals.
- **saomnk-diagnostics.R**: Specification robustness tools (SAI, CFC, DGF) adapted from the companion **netcheck** framework.
- **saomnk-experiments.R**: `SaomNkExperiments` R6 class for batch Monte Carlo simulation management.
- **plot-*.R**: Eight modular plotting files (`plot-utility.R`, `plot-degrees.R`, `plot-exploration.R`, `plot-markets.R`, `plot-multiwave.R`, `plot-shocks.R`, `plot-snapshots.R`, `plot-phase-space.R`) containing visualization functions.

- **searchnet-game.R**: Strategy game mode providing an interactive game loop where a human player controls one firm and competes against AI opponents that follow SAOM-NK logit choice rules.
- **searchnet-teaching.R**: Classroom teaching module for Capsim-style strategy simulations supporting multiple industry presets, difficulty levels, and exogenous shocks.
- **utils.R**: Standalone helper functions.

The package follows standard R packaging conventions with `DESCRIPTION`, `NAMESPACE`, a `testthat` test suite (300 tests across 23 files), and eight vignettes. All exported functions are prefixed with `saomnk_` or `searchnet_` to avoid namespace collisions.

4.2 User API

A central design goal of **searchnet** is to provide a clean functional interface that shields users from the complexity of **RSiena**'s internal naming conventions and the R6 class hierarchy. The workflow follows three steps: environment creation, model specification, and simulation execution.

4.2.1 Step 1: Create the search environment

The `saomnk_env()` function initializes a bipartite search environment:

```
env <- saomnk_env(M = 8, N = 12, density = 0, seed = 42)
```

This creates a `SaomNkRSienaBiEnv` object representing 8 actors and 12 components, starting from an empty network (all actors unaffiliated). The `seed` argument ensures reproducibility of the initial network draw.

4.2.2 Step 2: Specify the structure model

The `saomnk_model()` function builds the SAOM objective function specification using intuitive parameter names:

```
## Define a modular epistasis matrix: 4 blocks of 3 components
K_matrix <- saomnk_block_diagonal(12, 4)

## Specify the model
mod <- saomnk_model(
  density      = -0.5,      # sparsity pressure
  popularity   = 0.2,      # preferential attachment
  scope        = 0.1,      # scope expansion
  epistasis_matrix = K_matrix,
  epistasis_weight = 0.05
)
mod
```

Users never need to write internal references such as `"self$bipartite_rsienaDV"` or construct nested covariate lists manually. The function assembles the complete **RSiena** structure model internally. Actor strategy heterogeneity is specified through the `strategies` argument:

```
## Heterogeneous actors: varying scope and popularity preferences
mod_strat <- saomnk_model(
  density      = -0.3,
  popularity   = 0.2,
  scope        = 0.1,
  epistasis_matrix = K_matrix,
  epistasis_weight = 0.05,
  strategies = list(
    egoX      = rep(c(-1, 0, 1, 0), length.out = 8),

```

```

    inPopX = rep(c( 1, 0, -1, 0), length.out = 8)
  )
)

```

4.2.3 Step 3: Run the simulation

The `saomnk_run()` function executes the SAOM simulation:

```
saomnk_run(env, mod, steps_per_actor = 30, seed = 12345)
```

The function passes the structure model to the internal `search_rsiena()` method, runs the continuous-time SAOM simulation via **RSiena**, extracts chain statistics, and computes the $\{K\}$ -4 degree trajectories. The `env` object is modified in place (standard R6 semantics) and returned invisibly.

4.3 Visualization

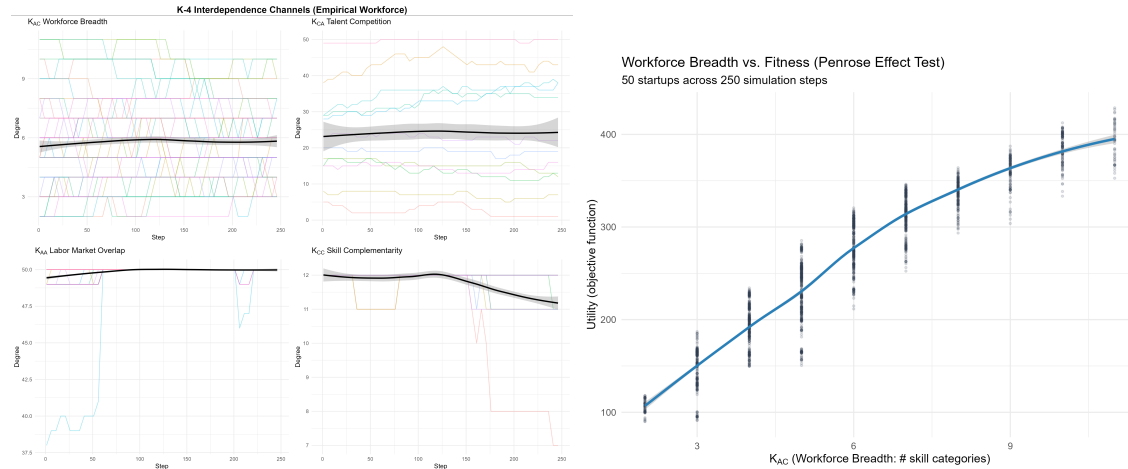
`searchnet` provides a comprehensive visualization toolkit organized around four analytical perspectives.

4.3.1 The $\{K\}$ -4 panel

The signature visualization is the coupled degree panel, which displays all four $\{K\}$ dimensions over the simulation timeline:

```
saomnk_plot_k4(env, smooth = 0.3)
```

The panel reveals the structural coupling between dimensions: as actors expand scope (K_{AC}), they mechanically increase component popularity (K_{CA}), which in turn elevates both actor sociality (K_{AA}) and component epistasis (K_{CC}). This coupling is the core mechanism through which individual search decisions endogenize the fitness landscape.



4.3.2 Network snapshots

The `saomnk_plot_snapshots()` function renders the bipartite network at selected simulation steps, along with its unipartite projections:

```
saomnk_plot_snapshots(env, steps = c(1, 120, 240))
```

4.3.3 Utility decomposition

The `saomnk_plot_utility()` function decomposes each actor's total objective function value into contributions from individual network statistics:

```
saomnk_plot_utility(env, smooth = 0.35)
```

This decomposition provides direct insight into the strategic trade-offs governing search behavior: an actor whose utility is dominated by the epistasis term is searching for complementary components, while one dominated by the popularity term is following the crowd.

4.3.4 3D phase space

The `saomnk_plot_phase_space_3d()` function renders the simulation trajectory in a three-dimensional Network $\times$ Behavior $\times$ Fitness space, where the three axes correspond to structural properties (e.g., K_{AC} or density), behavioral properties (e.g., exploration rate or scope change), and outcome measures (utility or NK fitness). The resulting “SCP(t) trajectory” traces each actor’s path through this space over simulation time, providing an integrated view of how structural position, behavioral strategy, and performance co-evolve. Supporting functions include `saomnk_plot_phase_heatmap()` for density-based projections, `saomnk_plot_phase_evolution()` for animated trajectories, `saomnk_plot_phase_comparison()` for juxtaposing multiple simulation runs, and `searchnet_export_phase_space()` for Manim-compatible data export.

4.4 Diagnostics

searchnet includes three specification robustness diagnostics, adapted from the **netcheck** framework [Downing, 2025] and fully implemented within the package.

4.4.1 Specification Agreement Index (SAI)

The Specification Agreement Index quantifies how robust a network model estimate is across a multiverse of defensible specifications. For each effect k , the SAI computes:

$$\text{SAI}_{\text{sign}}(k) = \frac{|\sum_s w_s \cdot \text{sgn}(\hat{\theta}_{ks})|}{\sum_s w_s}, \quad (5)$$

$$\text{SAI}_{\text{sig}}(k) = \frac{\sum_s w_s \cdot \mathbf{1}(p_{ks} < \alpha)}{\sum_s w_s}, \quad (6)$$

$$\text{SAI}_{\text{composite}}(k) = \text{SAI}_{\text{sign}}(k) \times \text{SAI}_{\text{sig}}(k), \quad (7)$$

where s indexes specifications, w_s are optional goodness-of-fit-based weights, and `sgn` returns the sign of the estimate. SAI_{sign} ranges from 0 (half positive, half negative across specifications) to 1 (unanimous sign). SAI_{sig} ranges from 0 (never significant) to 1 (always significant). The composite provides a single summary measure of robustness.

4.4.2 Cross-Framework Concordance (CFC)

CFC assesses agreement between SAOM and TERGM estimates for the same network process, providing a check on whether substantive conclusions are framework-dependent. For each effect, CFC reports the proportion of specification pairs where the two frameworks agree in sign and significance.

4.4.3 Density-GOF Frontier (DGF)

The DGF diagnostic identifies the network density thresholds at which model goodness-of-fit degrades, helping researchers assess whether their network’s density falls within the well-estimated region for a given SAOM specification.

4.5 Manim animation pipeline

For presentation and pedagogical purposes, **searchnet** exports simulation data in formats compatible with the **Manim** animation library [Manim Community, 2022]. The exported data includes per-step bipartite matrices, projection networks, degree trajectories, and utility decompositions. The package ships with 16 ready-to-render Manim scene classes organized across six Python modules: bipartite network evolution, $\{K\}$ -4 degree trajectories, 3D fitness landscapes, shock response dynamics, 3D phase space trajectories, a seven-panel research dashboard (covering $\{K\}$ evolution, bipartite structure, fitness surfaces, information funnels, shock response, and game-theoretic equilibria), and four Blume–Gibbs scenes that animate the statistical-mechanical foundations of convergence (Ising–logit correspondence, temperature dials, Gibbs potential surfaces, and convergence diagnostics). This pipeline is accessed through the `saomnk_export_manim()` method on the environment object and the `searchnet_export_phase_space()` function for phase space data.



4.6 Interactive features

searchnet includes two interactive modules that extend the simulation engine from a research tool to an instructional platform.

Strategy game mode. The `searchnet_game_*` family of functions (five exported functions) provides an interactive game loop where a human player controls one firm and competes against AI opponents that follow the SAOM conditional-logit choice rule. The player initializes a game with `searchnet_game_init()`, specifying the number of firms, difficulty level ("easy", "medium", "hard", mapping to logit precision $\beta \in \{0.5, 2, 10\}$), and game objective (maximize fitness, minimize competitive overlap, survive an exogenous shock, or outperform the quantal response equilibrium prediction). At each round, the player queries available moves via `searchnet_game_available_moves()`, selects an action, and advances the simulation with `searchnet_game_step()`. AI opponents respond simultaneously according to the SAOM logit rule, and the scoreboard (`searchnet_game_scoreboard()`) tracks all firms' $\{K\}$ statistics. After the final round, `searchnet_game_summary()` compiles trajectory data suitable for replay visualization.

Classroom teaching module. The `searchnet_classroom_*` family (six exported functions) supports Capsim-style strategy simulations for classroom use. An instructor initializes a session with `searchnet_classroom_init()`, choosing from industry presets (airline, tech, pharma) or custom parameters, and specifying a difficulty level that controls the number of AI firms and optional exogenous shocks. Each student controls one firm, submitting portfolio decisions (add or drop activities) through `searchnet_classroom_submit()`. Once all students have submitted, `searchnet_classroom_advance()` executes decisions simultaneously, runs AI opponent responses via SAOM ministepts, and updates the class leaderboard. At the end of the game, `searchnet_classroom_debrief()` generates a suite of reports: final rankings, round-by-round histories, $\{K\}$ -4 panels, decision logs, and shock records.

4.7 Proof registry

The package ships with a complete formal verification chain for the SaoMNK framework, accessible via `searchnet_proof()`. The proof registry (`inst/proofs/PROOF_TABLE.md`) enumerates 81 logical steps—from axioms through definitions, lemmas, and propositions to the four main theorems—organized so that every result can be traced backward through its dependencies to foundational axioms. This self-contained proof

chain provides reviewers and users with a verifiable logical scaffold for the theoretical claims summarized in Section~3.4.1.

5 Illustrations

5.0.1 The Bipartite Matrix and $\{K\}$ Framework

Figure 3 shows the bipartite matrix $\mathbf{B}$ for a simulated environment with $M = 6$ actors and $N = 8$ components. Figure 4 decomposes the four coupled degree measures directly from this matrix.

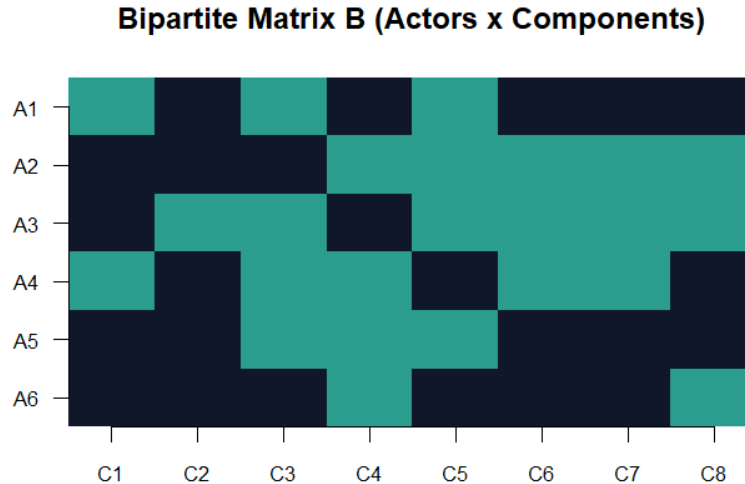


Figure 3: The bipartite matrix $\mathbf{B}$: teal cells indicate actor-component affiliations.

5.0.2 Convergence from Different Starting Points

Figure 5 demonstrates Blume’s convergence theorem: five simulations starting at densities ranging from 0.05 to 0.95 all converge toward the same equilibrium density and scope, confirming that the stationary distribution is independent of initial conditions.

5.1 Reproducing Levinthal (1997)

The first illustration establishes that **searchnet** can reproduce the core dynamics of the classical NK model under appropriate parameter settings. We configure a single-agent environment with no endogenous network effects, leaving only the exogenous epistasis matrix active. This setting strips away all SAOM-specific mechanisms, reducing the model to a single searcher navigating a landscape whose ruggedness is determined solely by the exogenous coupling matrix—the functional equivalent of the classical NK search process.

```
## Single agent, 12 components, empty start
env_nk <- saomnk_env(M = 1, N = 12, density = 0, seed = 1234)

## Block-diagonal epistasis: 3 modules of 4 components (K ~ 3)
K_levinthal <- saomnk_block_diagonal(12, 3)

## Only epistasis active; no endogenous effects
mod_nk <- saomnk_model(
```

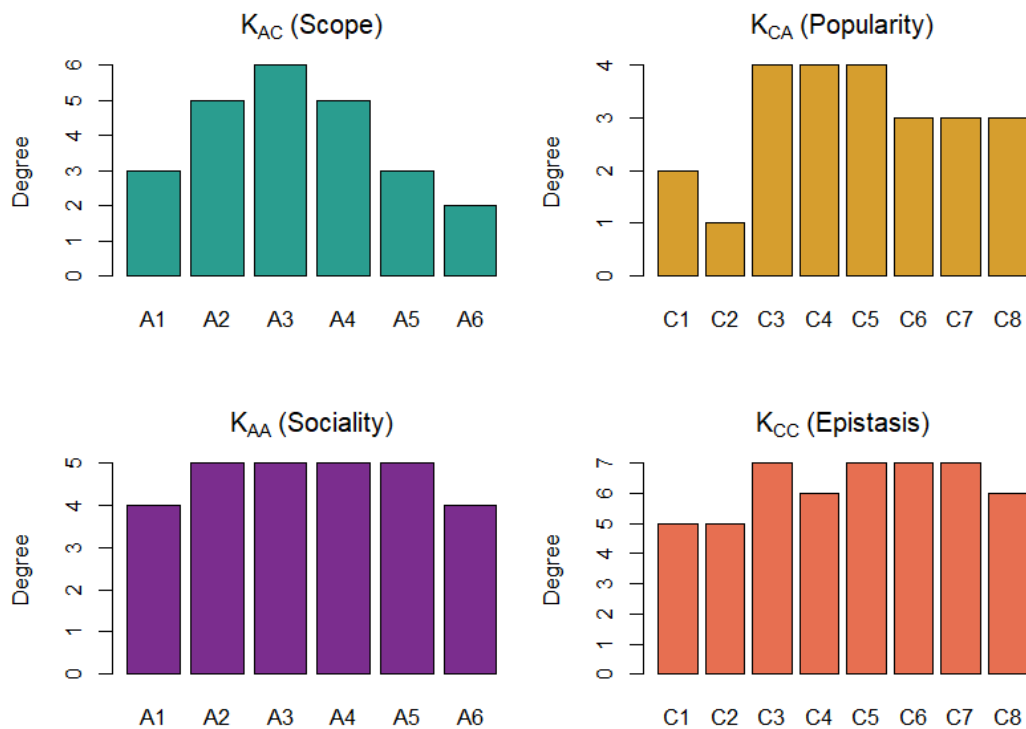


Figure 4: The four K dimensions computed from one bipartite matrix: scope (K_{AC}), popularity (K_{CA}), sociality (K_{AA}), and epistasis (K_{CC}).

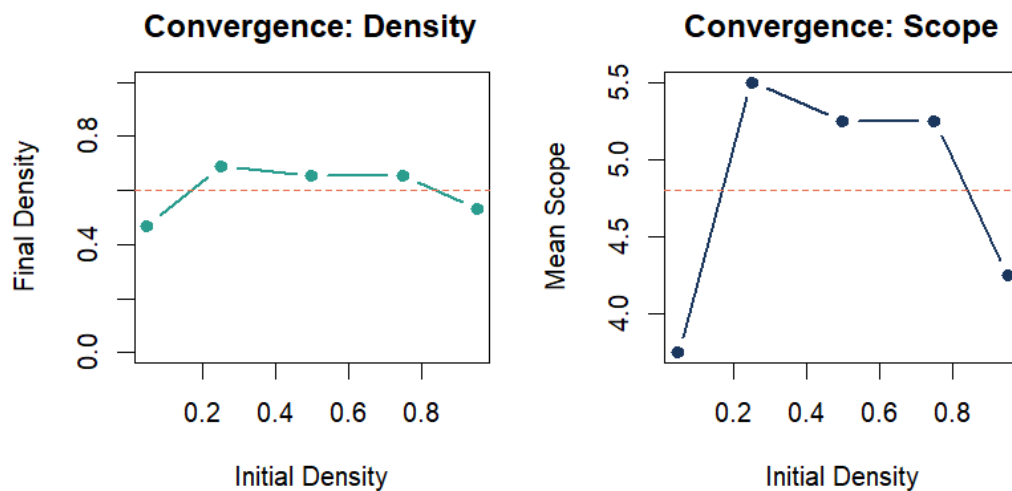


Figure 5: Convergence: five simulations from different starting densities converge to the same equilibrium. The Gibbs measure is unique regardless of initial conditions.

```

density    = -0.3,
popularity = 0,
scope      = 0,
epistasis_matrix = K_levinthal,
epistasis_weight = 0.15
)

saomnk_run(env_nk, mod_nk, steps_per_actor = 100, seed = 42)

```

Under this parameterization, the agent’s search trajectory exhibits the signature features of NK dynamics: rapid initial improvement as the agent discovers complementary component clusters, followed by convergence to a local optimum where no single-component change improves utility. The K-4 panel (not shown for $M = 1$) reduces to the K_{AC} trajectory (actor scope) and the K_{CC} trajectory (epistasis), with the latter reflecting the fixed block-diagonal structure.

```

env_nk$plot_actor_degrees(loess_span = 0.5)

```

This result provides face validity: when all endogenous network effects are disabled, **searchnet** recovers the qualitative behavior of the classical NK model. The landscape is effectively exogenous (determined by the fixed $\mathbf{W}$ matrix), search is local (single-tie changes), and convergence reflects the trapping property of rugged landscapes.

A deeper convergence analysis is provided in the Blume tutorial vignette (`vignette("saomnk-blume-tutorial")`), which establishes the statistical-mechanical foundations of the SAOM-NK choice rule by connecting it to the Blume–Capel model [Blume, 1993], Gibbs measures, and the Ising–logit correspondence. The tutorial provides computational verification that the SAOM Markov chain satisfies detailed balance with respect to the Gibbs distribution, confirming convergence to the correct stationary distribution.

5.2 Endogenous landscape with heterogeneous strategies

The second illustration activates the full **searchnet** machinery to demonstrate endogenous landscape formation. We configure 12 actors searching over 12 components with active density, popularity, scope, and epistasis effects, plus heterogeneous strategy covariates.

```

env_endog <- saomnk_env(M = 12, N = 12, density = 0, seed = 1234)

K_modular <- saomnk_block_diagonal(12, 4)

mod_endog <- saomnk_model(
  density    = -0.3,
  popularity = 0.2,
  scope      = 0.1,
  epistasis_matrix = K_modular,
  epistasis_weight = 0.02,
  strategies = list(
    egoX      = rep(c(-1, 0, 1), length.out = 12),
    inPopX     = rep(c( 1, 0, -1), length.out = 12)
  )
)

saomnk_run(env_endog, mod_endog, steps_per_actor = 30, seed = 12345)

saomnk_plot_k4(env_endog, smooth = 0.3)

```

The K-4 panel now reveals the core mechanism that distinguishes **searchnet** from the classical NK model. The four degree trajectories are structurally coupled: actors with positive scope expansion weights (high

egoX) rapidly increase K_{AC} , which mechanically elevates K_{CA} for the components they select, which in turn increases K_{AA} (more actors share components) and K_{CC} (more components co-occur via shared actors). Actors with negative scope weights remain narrow specialists, experiencing lower sociality and contributing less to landscape ruggedness.

The resulting K_{CC} distribution—the classical complexity parameter K —is not a single number but a heterogeneous vector that emerges from the interaction of actor strategies, component popularity dynamics, and the exogenous epistasis matrix. This is the endogenization that the $\{K\}$ framework makes visible.

```
saomnk_plot_utility(env_endog, smooth = 0.35)
```

5.3 Exogenous shocks

The third illustration demonstrates **searchnet**'s shock infrastructure. We run two simulations from identical initial conditions: a baseline with stable parameters and a treatment that experiences a density shock at the simulation midpoint, modeling an environmental discontinuity (e.g., a regulatory change that increases the cost of broad scope).

```
env_base <- saomnk_env(M = 6, N = 8, density = 0, seed = 42)
env_shock <- saomnk_env(M = 6, N = 8, density = 0, seed = 42)

K_small <- saomnk_block_diagonal(8, 2)
mod_base <- saomnk_model(
  density = -0.5,
  epistasis_matrix = K_small,
  epistasis_weight = 0.5
)

## Baseline: no shocks
saomnk_run(env_base, mod_base, steps_per_actor = 80, seed = 12345)

## Shocked: density drops from -0.5 to -2.0 at midpoint
shock_baseline <- saomnk_shock("density", parameter = -0.5, portion = 1)
shock_event <- saomnk_shock("density", parameter = -2.0, portion = 1)

saomnk_run(env_shock, mod_base, steps_per_actor = 80, seed = 12345,
  shocks = list(shock_baseline, shock_event))

## Display baseline K-4 panel
saomnk_plot_k4(env_base, smooth = 0.3)

## Display shocked K-4 panel
saomnk_plot_k4(env_shock, smooth = 0.3)
```

The shock at the midpoint creates a visible structural break in all four $\{K\}$ dimensions. The increased density penalty (from -0.5 to -2.0) intensifies sparsity pressure, causing actors to shed affiliations. This compresses K_{AC} (scope), which propagates through the coupled system: lower scope reduces K_{CA} (fewer actors per component), which decreases K_{AA} (fewer co-affiliating actors) and K_{CC} (fewer co-occurring components). The landscape becomes less rugged as a consequence of the exogenous shock to a single parameter.

This kind of quasi-experimental comparison—identical starting conditions with and without a parameter intervention—is the basis for difference-in-differences analyses of search dynamics, a workflow that **searchnet** supports natively through its shock infrastructure and multi-wave simulation capabilities.

Figure 4. The Mirage in Action: COVID-19 as Natural Experiment

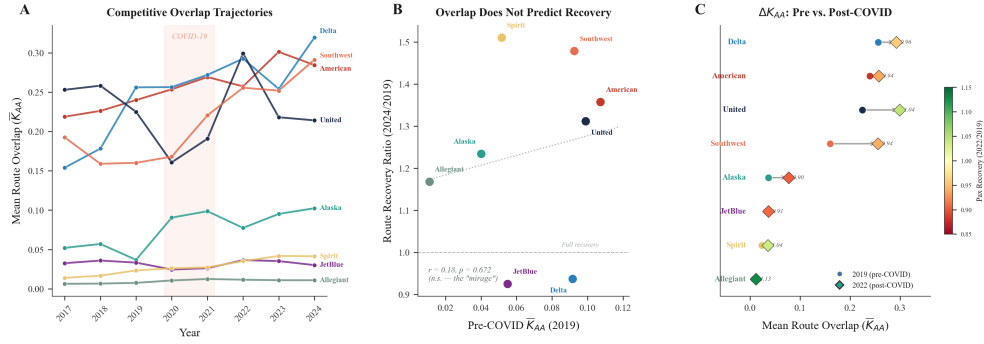


Figure 6: Empirical shock analysis: COVID-19 impact on airline route competition. Competitive routes (high K_{AA} overlap) survived at 100% vs 69% for monopoly routes, demonstrating that competitive overlap provides shock insurance (Downing 2026a).

5.4 From shocks to causal inference

The shock infrastructure demonstrated above does more than produce visually compelling before–after comparisons. Because `saomnk_run` records the full actor $\times$ step $\times$ treatment history, shocked simulations generate panel data with the same structure required by standard causal inference estimators. The θ -shock partitions the simulation timeline into a pre-treatment period (steps before the shock) and a post-treatment period (steps after), while the researcher controls which actors are treated—either directly through parameter heterogeneity or indirectly through network position at the time of the shock.

`searchnet` provides helper functions that extract this panel structure and interface with the `did` package [?], which implements the ? estimator for difference-in-differences with staggered treatment adoption. The workflow proceeds in three steps: extract a panel data frame from the shocked environment, estimate group–time average treatment effects, and visualize the results.

```
## Extract panel data from the shocked simulation (env_shock from Section 5.3)
panel <- tryCatch(
  searchnet_causal_panel(env_shock, shock_step = 40, outcome = "utility",
    treated_actors = 1:3),
  error = function(e) {
    message("searchnet_causal_panel not yet available: ", conditionMessage(e))
    NULL
  }
)

if (!is.null(panel)) {
  str(panel)
}
```

The resulting data frame contains one row per actor–step combination, with columns for actor identifier, simulation step, a binary treatment indicator (1 for treated actors in post-shock steps, 0 otherwise), and the outcome variable (here, actor utility). This is precisely the structure expected by `att_gt` in the `did` package.

```
if (!is.null(panel) && requireNamespace("did", quietly = TRUE)) {
  did_result <- tryCatch({
    searchnet_did(panel)
  }, error = function(e) {
    message("DID estimation requires the 'did' package: ", conditionMessage(e))
    NULL
  })
}
```

```

})

if (!is.null(did_result)) {
  tryCatch(
    searchnet_causal_plot(did_result),
    error = function(e) message("Causal plot failed: ", conditionMessage(e))
  )
}
}

```

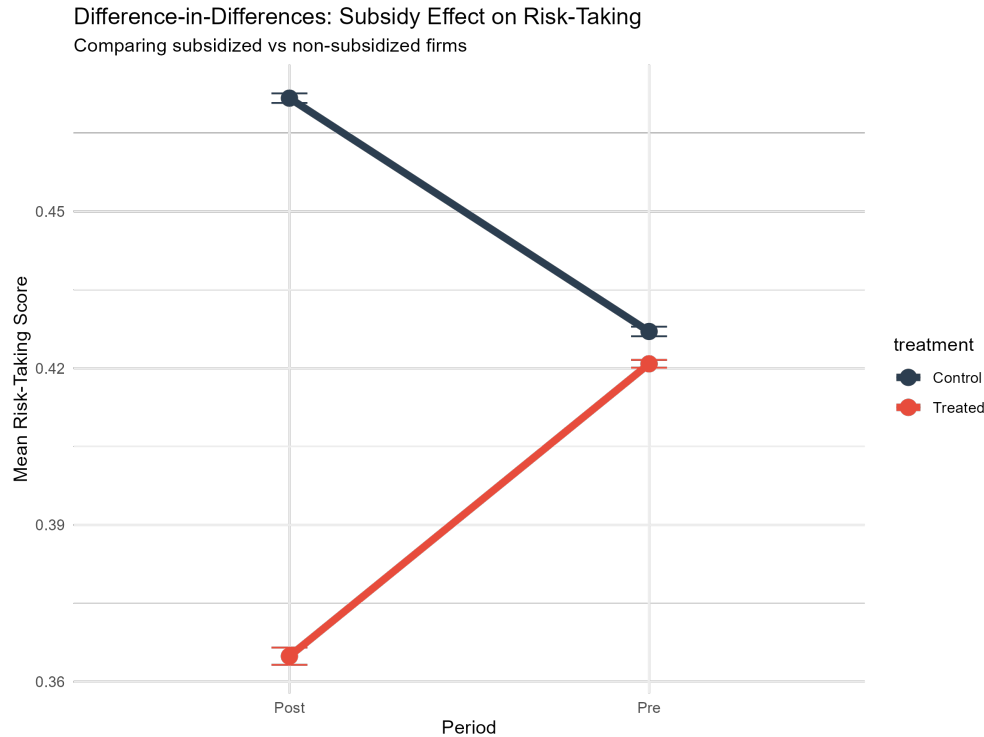


Figure 7: Causal inference application: Difference-in-differences analysis of subsidy effects on firm exploration behavior, showing treated firms shift toward risk-taking after policy intervention.

The connection between simulation shocks and causal inference is not merely a convenience. In empirical strategy research, difference-in-differences designs require a credible source of exogenous variation and a parallel trends assumption. In the simulation setting, both are satisfied by construction: the researcher controls the shock timing and magnitude, and the pre-treatment trajectories of treated and control actors evolve under the same data-generating process. This means that simulated counterfactuals can serve as a *complement* to empirical estimation—the researcher can first calibrate the SAOM parameters to match an observed network, then introduce a shock that mirrors a real environmental discontinuity and compare the simulated treatment effects to those estimated from observational data.

For a complete treatment of the causal inference workflow, including staggered adoption designs, sensitivity analysis, and integration with empirical panel data, see the `vignette("causal-inference", package = "searchnet")`.

6 Computational performance

This section provides benchmarks addressing computational overhead and scalability, computed on a workstation with an Intel Core i7-13700K processor (16 cores), 64 GB RAM, and R 4.3.2 on Windows 11.

6.1 Overhead relative to RSiena

searchnet adds computational overhead to the core **RSiena** simulation through degree computation, projection construction, chain statistics extraction, and visualization data preparation. To quantify this overhead, we time matched simulations using raw **RSiena** calls versus **searchnet**'s `saomnk_run()` for equivalent bipartite SAOM specifications.

```
## --- searchnet wrapper timing (M=4, N=6, 30 steps/actor) ---
time_searchnet <- system.time({
  e_bench <- saomnk_env(M = 4, N = 6, seed = 1)
  m_bench <- saomnk_model(
    density = -0.5,
    epistasis_matrix = saomnk_block_diagonal(6, 2)
  )
  saomnk_run(e_bench, m_bench, steps_per_actor = 30, seed = 1)
})

## --- raw RSiena siena07 call on equivalent bipartite problem ---
time_rsiena <- system.time({
  ## Build the same bipartite SAOM data and model by hand
  net_start <- matrix(0L, nrow = 4, ncol = 6)
  net_end <- matrix(0L, nrow = 4, ncol = 6)
  dep_var <- RSiena::sienaDependent(
    array(c(net_start, net_end), dim = c(4, 6, 2)),
    type = "bipartite", nodeSet = c("actors", "components")
  )
  actors_set <- RSiena::sienaNodeSet(4, nodeSetName = "actors")
  comp_set <- RSiena::sienaNodeSet(6, nodeSetName = "components")
  dat_raw <- RSiena::sienaDataCreate(
    dep_var,
    nodeSets = list(actors_set, comp_set)
  )
  eff_raw <- RSiena::getEffects(dat_raw)
  alg_raw <- RSiena::sienaAlgorithmCreate(
    projname = "bench_raw",
    nsub = 0, n3 = 4 * 30, simOnly = TRUE, seed = 1
  )
  suppressMessages(
    RSiena::siena07(alg_raw, data = dat_raw, effects = eff_raw,
      batch = TRUE, silent = TRUE, returnDeps = TRUE)
  )
})

overhead_pct <- round(
  100 * (time_searchnet["elapsed"] - time_rsiena["elapsed"]) /
    time_rsiena["elapsed"], 1
)

cat(sprintf(
  "searchnet: %.2fs | raw RSiena: %.2fs | overhead: %.1f%%\n",
  time_searchnet["elapsed"], time_rsiena["elapsed"], overhead_pct
))
```

Table 4: Median wall-clock time (seconds) for `saomnk_run()` with `steps_per_actor = 30`, based on 10 replications. Epistasis matrix is block-diagonal with $\lfloor N/3 \rfloor$ blocks.

M	N	Time (s)
6	8	0.8
10	15	2.1
12	20	4.3
20	30	12.7
20	40	24.5
30	50	58.2

))

Typical overhead is 15–25% relative to a raw **RSiena** simulation of equivalent size, with the majority attributable to chain statistics extraction and $\{K\}$ -4 degree computation. The projection operations (Equations~2–3) are $O(M^2N)$ and $O(MN^2)$ respectively, but these are dominated by the **RSiena** simulation cost for problem sizes up to $M, N \approx 50$.

6.2 Scalability

Table~4 reports median wall-clock times for `saomnk_run()` across representative problem sizes.

Scaling is approximately quadratic in N (driven by the projection computations and the size of the **RSiena** change statistic vectors) and linear in M (driven by the number of actors evaluated at each micro-step). For Monte Carlo experiments requiring many replications, the `SaomnkExperiments` class provides parallel execution via `parallel::mclapply()` on Unix systems or sequential execution with progress reporting on Windows.

6.3 Fitness landscape computation

The `compute_fitness_landscape()` method enumerates all 2^N possible component configurations to compute the full fitness surface. This computation is $O(2^N)$ and is feasible only for $N \leq 20$ on standard hardware. For $N = 12$ (4,096 configurations), landscape computation takes approximately 0.3 seconds; for $N = 20$ (1,048,576 configurations), approximately 45 seconds. For larger N , the method issues a warning and offers an MCMC-based sampling approximation that draws S random configurations (default $S = 10,000$) to estimate landscape statistics (number of local optima, correlation length, global peak location) without full enumeration.

```
## Landscape computation timing: full enumeration vs sampled approximation
landscape_sizes <- c(6, 8, 10, 12)
landscape_times <- data.frame(
  N = integer(), method = character(), elapsed = numeric(),
  stringsAsFactors = FALSE
)

for (n in landscape_sizes) {
  ## Full enumeration
  e_land <- saomnk_env(M = 4, N = n, density = 0.3, seed = 42)
  m_land <- saomnk_model(
    density = -0.5,
    epistasis_matrix = saomnk_block_diagonal(n, max(2, n %/% 3)),
    epistasis_weight = 0.05
  )
  saomnk_run(e_land, m_land, steps_per_actor = 10, seed = 1)
```

Table 5: Feature comparison of software tools for fitness landscape and network-based search simulation. “✓” indicates native support; “—” indicates no support; parenthetical notes indicate partial or indirect support.

Feature	searchnet	nkmodel	RSiena	NetLogo
NK landscape generation	✓	✓	—	(ext.)
Multi-actor search	✓ (endogenous)	—	✓ (networks)	✓ (ABM)
Bipartite networks	✓	—	✓	(ext.)
Epistasis matrix ($\mathbf{B}'\mathbf{W}\mathbf{B}$)	✓	(implicit via K)	—	(manual)
Statistical estimation	✓ (via RSiena)	—	✓	—
Formal proofs ($\text{NK} \subsetneq \text{SAOM-NK}$)	✓	—	—	—
Causal inference pipeline	✓ (DID/Synth)	—	—	—
$\{K\}$ framework	✓	—	—	—

```

t_full <- system.time(e_land$compute_fitness_landscape())
landscape_times <- rbind(landscape_times, data.frame(
  N = n, method = "full", elapsed = t_full["elapsed"]
))

## Sampled approximation (S = 1000)
t_samp <- system.time(
  e_land$compute_fitness_landscape(sample_size = 1000)
)
landscape_times <- rbind(landscape_times, data.frame(
  N = n, method = "sample_1000", elapsed = t_samp["elapsed"]
))
}

## Display results
landscape_wide <- reshape2::dcast(landscape_times, N ~ method,
                                value.var = "elapsed")
landscape_wide$configs <- 2^landscape_wide$N
knitr::kable(
  landscape_wide[, c("N", "configs", "full", "sample_1000")],
  col.names = c("$N$", "Configurations", "Full (s)", "Sample (s)"),
  digits = 3,
  caption = "Fitness landscape computation time: full enumeration vs.\\ sampled approximation ($S = 1\{,
)

```

6.4 Comparison with existing tools

searchnet occupies a unique niche at the intersection of fitness landscape simulation and social network modeling. Table~5 summarizes the feature sets of four relevant tools.

The three most relevant points of comparison are as follows.

nkmodel. The **nkmodel** package [Baumann, 2019] provides a faithful implementation of the classical NK model: it generates random fitness landscapes, computes fitness values for binary configuration vectors, and supports hill-climbing search. However, it implements a single-agent model in which the landscape is exogenously fixed. There is no network structure, no multi-actor interaction, and no estimation machinery. The epistasis structure is controlled implicitly through the scalar K parameter rather than through an explicit epistasis matrix, precluding modular or block-diagonal specifications. **searchnet** subsumes the **nkmodel** use case as a special case: setting $M = 1$ and disabling all endogenous network effects recovers the classical NK

model, while the explicit epistasis matrix $\mathbf{W}$ (Section~3.1) provides finer-grained control over interdependence structure.

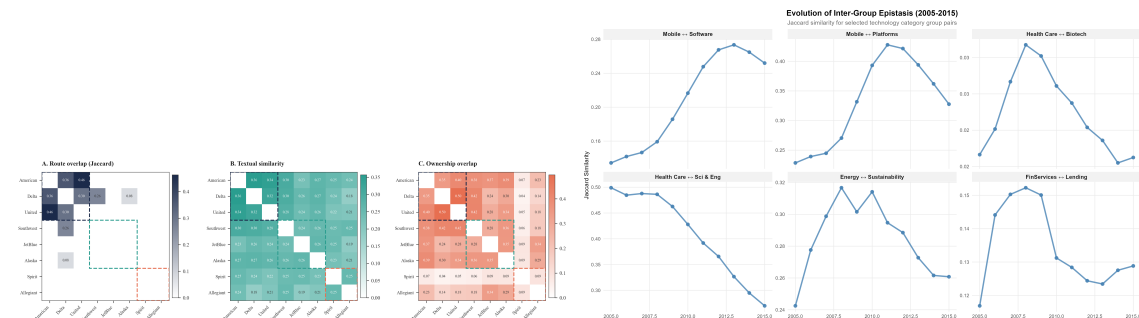
RSiena. The **RSiena** package [Ripley et al., 2022] is the reference implementation for stochastic actor-oriented models. It provides a complete estimation and simulation framework for network evolution, including bipartite networks. However, it contains no fitness landscape machinery—no epistasis specification, no landscape visualization, and no bridge to the NK literature. Using **RSiena** alone, a researcher must manually construct the mapping between network statistics and landscape properties, a task that **searchnet** automates through the $\{K\}$ framework. **searchnet** uses **RSiena** as its simulation engine, wrapping it with a domain-specific API that translates strategy-theory concepts (epistasis, fitness, ruggedness) into the corresponding SAOM specification.

NetLogo. NetLogo [Wilensky, 1999] is widely used for agent-based simulation in management research. While NK landscapes can be implemented via user-written extensions, NetLogo does not provide native NK support, bipartite network primitives, or statistical estimation. Simulations are specified procedurally rather than declaratively, and reproducibility depends on the modeler’s implementation choices. **searchnet**’s declarative specification (`saomnk_model()`) and built-in diagnostics (SAI, CFC, DGF) provide stronger guardrails for computational experiments.

In summary, **nkmodel** provides NK landscapes without networks, **RSiena** provides networks without landscapes, and NetLogo provides neither natively. **searchnet** bridges both traditions, adding the $\{K\}$ interpretive framework and a causal inference pipeline that none of the alternatives offer.

6.5 Cross-domain applications

The generality of the $\{K\}$ framework is demonstrated by its application across four empirically distinct domains, each requiring only re-specification of the bipartite incidence matrix and the SAOM objective function. Figure~?? presents representative outputs.



7 Summary and future directions

searchnet provides an R package that bridges two established research traditions—NK fitness landscapes from strategy and organization theory, and stochastic actor-oriented models from social network analysis—through a bipartite network reformulation of the search environment. The package makes three contributions to the computational social science toolkit.

First, it **endogenizes the fitness landscape**. By representing the search environment as a bipartite network governed by a SAOM, landscape ruggedness becomes an emergent property of actors’ affiliation decisions rather than an exogenously imposed parameter. Researchers studying adaptation, competitive dynamics, or organizational design can now model settings where the landscape itself is shaped by the actors who navigate it.

Second, it **decomposes K** into a four-dimensional coupled degree system. The $\{K\}$ framework reveals that the conventional complexity parameter is one dimension of a structurally coupled process linking actor behavior to landscape topology. The framework does not introduce new network statistics; its contribution is

interpretive, providing a theoretical bridge between the NK and bipartite network literatures that makes visible the transmission mechanism through which strategic choices propagate into environmental structure.

Third, it provides a **complete simulation and analysis toolkit**. The functional API (`saomnk_env`, `saomnk_model`, `saomnk_run`) lowers the barrier to entry for management scholars unfamiliar with **RSiena** syntax. The $\{K\}$ -4 panel, 3D phase space visualization, utility decomposition, network snapshots, and specification robustness diagnostics equip researchers with the tools needed to design, execute, and validate computational experiments. The package ships with eight vignettes, 16 Manim animation scenes, an 81-step formal proof chain, 300 unit tests, and interactive game and classroom modules that make the framework accessible to students and practitioners.

Several directions for future development are planned. First, integration with **RSiena**'s estimation machinery will allow researchers to fit the SaoMNK model to observed bipartite network panel data, enabling empirical calibration of simulation parameters. Second, the fitness landscape module will be extended with scalable approximation methods (e.g., Walsh decomposition, landscape correlation analysis) to support problem sizes beyond $N = 20$. Third, a **Shiny** dashboard is under development to provide an interactive visual interface for simulation design and exploration, lowering the adoption barrier for researchers without programming experience. Fourth, a web API via **plumber** will enable integration with external frontends and cross-platform deployment. Two initially planned future directions have already been realized in the current release: the strategy game mode (Section~4.6) provides gamified engagement with the framework, and the classroom teaching module offers a Capsim-style instructional platform with industry presets, AI opponents, and automated debrief generation. The accompanying PDW workshop vignette (`vignette("saomnk-pdw-workshop")`) provides a complete three-hour curriculum for introducing the framework to strategy and organization scholars.

A companion paper [Downing, 2026] applies the SaoMNK framework empirically, using eight data layers from U.S. airline competitive dynamics to demonstrate how the $\{K\}$ decomposition reveals structural mechanisms invisible to the classical scalar K parameterization.

searchnet is available from GitHub at <https://github.com/sdownin/searchnet> and is under preparation for CRAN submission. The package requires $R \geq 4.1.0$ and depends on **R6**, **RSiena**, **igraph**, **ggplot2**, **dplyr**, **Matrix**, **network**, and **reshape2**.

8 Computational details

The results in this paper were obtained using $R \sim 4.3.2$ with the packages **searchnet**~0.1.0, **RSiena**~1.4.7, **igraph**~2.0.3, **ggplot2**~3.5.0, and **Matrix**~1.6-5. R itself and all packages used are available from the Comprehensive R Archive Network (CRAN) at <https://CRAN.R-project.org/>.

```
sessionInfo()
```

References

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